

Shen (Sam) Xie

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EDUCATION

- **Georgia Institute of Technology, Atlanta, United States** (Aug 2026 – Present).
Master's, Major in Computational Science and Engineering; H. Milton Stewart School of Industrial and Systems Engineering.
- **Wuhan University, Wuhan, China** (Sep 2020 – Jun 2024).
Bachelor's, Major in Politics, Philosophy, and Economics; Minor in Computer Science and Engineering; Hongyi Honor College.
Honor Graduate; First Prize Academic Scholarship (ranked 2nd).

RESEARCH INTERESTS

- **AI for Scientific Discovery, Multimodal Learning, Recommender Systems, Human-AI Interaction, Digital Platforms, UGC.**

RESEARCH EXPERIENCE

- **HEC Paris, Paris, France** (Mar 2025 – Aug 2026).
Remote Research Assistant, Department of ISOM; supervisor: Prof. Xitong Li.
 - Conducting researches related to ML and HCI.
 - Built real-time data pipelines for **Reddit, SSRN, FastMoss, Tiktok Shop**, producing research datasets for creator-economy, platform consumer behavior, and digital marketing studies.
 - Designed influencer-product matching workflows with rule-based and machine-learning approaches.
- **Hong Kong University of Science and Technology, Hong Kong SAR, China** (Aug 2024 – Jul 2025).
Onsite Research Assistant, Department of ISOM; supervisor: Prof. Tatko Koh.
 - Reviewed literature on crowdsourcing contests, human-AI interaction, diversification, and platform strategy to support theory building and empirical design. Built a large-scale **99designs** dataset.
- **Wuhan University, Wuhan, China** (Feb 2024 – Jun 2024).
Research Assistant, Department of MSE; supervisor: Prof. Baojun Gao.
 - Conducted robustness checks and **Structural Topic Modeling** for an **Journal of Retailing (ABS 4)** paper under second review using TripAdvisor review data.

ONGOING RESEARCH PAPERS

- **ML: General-Purpose LLM Agents for Efficient Scientific Experiment Selection** (Sep 2026 – Present).
Co-first author with Yuming Zeng; Preparing for submission to ICML 2027.
 - Developing a general framework for **sequential scientific experiment selection**, where an agent adaptively chooses the most informative next experiment from a large candidate space to accomplish a scientific objective with as few experiments as possible.
- **ML: Diffusion-Based Image–Audio Alignment** (Aug 2026 – Present).
Third Author with Biyang Xin, Mingwei Sun, Shijie Zhang, and Xitong Li; In progress.
 - Develops a diffusion-based multimodal framework for learning aligned **image–audio representations**, enabling cross-modal correspondence and generation across visual and acoustic modalities.
- **ML: EEG-to-Speech Decoding via Coordinate-Aware Cross-Modal Representation Learning** (Apr 2026 – Present).
First Author with Mingwei Sun, Shijie Zhang, and Xitong Li; Preparing for submission to ICML 2027.
 - Develops a unified **EEG-to-speech-content** framework that maps variable-channel EEG signals to shared **MFCC** and **HuBERT** speech representations using coordinate-aware CNN–Conformer encoders.
 - Builds leakage-controlled multi-dataset pipelines with subject/content OOD splits, masked channel–time modeling, contrastive EEG–audio alignment, and counterfactual EEG controls for rigorous evaluation.

- **HCI: When VLM Frames the Product Advertising** (Nov 2025 – Aug 2026).
Second author. Under review at CHI 2027; Revising for Journal submission.
 - Details as well as the tiles withheld during peer review.
- **IS: The Power of Content Similarity in Recent Reviews on Hotel Revenue** (May 2025 – Present).
Second author with Baojun Gao and Jiawen Wang; In preparation to submit to International Journal of Hospitality Management (ABS 3); [GitHub](#).
 - Analyzes **1.5M TripAdvisor reviews** and revenue data from **2,990 hotels** across Texas cities.
 - Finds that higher adjacent review similarity significantly reduces revenue, especially for high-end, high-volume, and lower-rated hotels.

SELECTED GITHUB PROJECTS

- **Software Engineering: Influencer Agent for TikTok Shop** (Sep 2025 – Aug 2026).
with start-up: InFound.ai; [GitHub](#).
 - Led backend and experimentation development for an **AI-native influencer marketing system** for MacOS and Windows, covering creator search, profiling, automated outreach and chat, sample management, sales-data processing, and API / MySQL / service integration.
 - Built real-time processing and deployment-facing infrastructure that connects creator matching, recommendation delivery, and operational analytics into an end-to-end data loop from outreach to fulfillment.
 - Received the **Gold Award** at the Guangdong-Hong Kong-Macao Greater Bay Area Inno Awards Demo Day.
- **ML Engineering: Generative Ad Recommendation with RQ-VAE Semantic IDs and RankMixer Bidding** (Oct 2025 – Mar 2026).
with an advertisement company located in Shenzhen, China.
 - Learned **RQ-VAE**-based discrete semantic codes for joint image-text **5120-d embeddings**, compressing them into **4-d × 256-way Semantic IDs** with **over 92%** utilization in all 4 codebooks. [GitHub](#)
 - Designed a **RankMixer** bidding reranker with parameter-free token mixing, per-token FFN / **SparseMoE**, and joint **CTR+CTCVR** training, delivering offline **+0.5%** AUC and online lifts in participation, win, impression, and **PctCVR**. [GitHub](#)
- **Web Crawling: Social Science Papers Auto Crawler and Analysis Agent** (Apr 2025 – Jun 2025).
with Ziqing Li; [GitHub](#).
 - Built a crawling agent and dataset covering more than **1.2 million papers**, including titles, abstracts, author information, and citation links.
 - Designed the pipeline to support downstream topic discovery, citation analysis, and automated literature exploration.

SEMINAR OUTREACH

- Chinese Open Science Network Hackathon 2025 (Aug 2025). *Invited by Southern University of Science and Technology and University of Macau.*
- Quant Marketing PhD Summer Workshop (Jul 2025). *Invited by City University of Hong Kong.*
- AI for Business Summer Workshop (Jul 2025). *Invited by University of Sci and Tech of China; **Outstanding Participant Award**.*
- Paper Sharing: Unexpected Negative Effect from Platform Regulation (May 2025). *Management Sci Seminar co-hosted by Wuhan University and Xi'an JiaoTong University.*

SKILLS

- Programming languages: **Python, R, Stata, SQL, L^AT_EX**.
- Engineering stack: **FastAPI, Redis, RabbitMQ, Docker, PostgreSQL, MySQL**.
- Research and modeling: **recommender systems**, NLP, structural topic modeling, multimodal modeling, and large-scale web crawling.